

II-2 Prompt Reactivity Feedback Phenomena¹

Reactivity feedback is the phenomenon that occurs when an originally applied reactivity changes the state of the system, normally via a change in the neutron flux and subsequent effects. This change in the reactor causes a change in reactivity, $\delta\rho_{fb}$, which in turn feeds back to and modifies the original reactivity.

Reactivity feedback is called “*inherent*” if its occurrence is based on an unavoidable and thus totally reliable physical phenomenon. An example is the Doppler broadening of resonances that is directly associated with the fuel temperature. Doppler broadening of resonances automatically leads - through reduction in resonance self-shielding – to an increase in neutron absorption and thus to a negative and inherent reactivity feedback (note that resonance capture dominates resonance fission in all power reactors).

Such feedback is called “*prompt*” if it directly follows the fuel temperature changes. If additional physical phenomena such as material motion or heat transfer are required to produce a certain feedback effect, a delay exists between the energy production in the fuel and the feedback. The delay in axial fuel expansion is small enough that this expansion can also be considered as a prompt feedback effect.

Power transients under the influence of temperature-stimulated counteraction (feedback) have been investigated since the realization of the first chain reaction. Feedback, especially prompt and inherent reactivity feedback, is vitally important for the nuclear reactor safety. If a reactor should become superprompt critical, the neutron flux could – in the absence of feedback – rise so rapidly that mechanical shutdown devices could not reduce the power before it reached a destructive level. Fortunately, immediately counteracting shutdown phenomena that are *inherent* in the fuel rapidly reduces the superprompt-critical reactivity to subprompt-critical values. If the reactivity quickly returns to subprompt-critical values, a rapid decrease in the reactor power results. Consequently, the total energy release of such a transient can stay below damaging levels or cause only limited damage. Since the counteracting reactivity feedback is inherent, the resulting shutdown effect is physically inevitable and cannot fail to occur. The impossibility of failure distinguishes *inherent* phenomena from *man-made* shutdown devices; the latter have finite failure probabilities. The presence of strong inherent shutdown phenomena is extremely important for any commercial nuclear power production.

Prompt Feedback Reactivity

Fuel Temperature Rise

The rise of the fuel temperature during a transient is due to the energy directly produced in fission reactions, which amounts to ~ 200 MeV per fission. The directly produced fission energy appears in the form of two fission products. Small additional contributions are due to scattering and capture of fission neutrons. The γ quanta emitted along with neutron capture amount to ~ 5 to 7 MeV per capture (~ 10 MeV per fission). Most of the original energy of the fission neutrons is lost in scattering (~ 5 to 6 MeV per fission). A further contribution is due to the radioactive decay of fission products. For an accurate treatment of feedback during transients, these contributions

¹ Source:

1. Ott & Neuhold, *Introductory Nuclear Reactor Dynamics*, American Nuclear Society, 1985

should be subdivided into a part deposited in the fuel and a part deposited outside of the fuel material (cladding, coolant, moderator, and blanket or reflector).

The major part of the energy released in the fission process (~83%) appears as a kinetic energy of the two fission fragments. The fission fragments emerge as highly ionized atoms. They lose their energy through ionization within a very small range ($\leq 10^{-2}$ mm in the fuel). Ion pairs produced directly by the fission products also quickly dissipate their energy. Thus, after a negligibly short time ($\delta t \ll \Lambda$) the energy of the fission fragments are converted into a temperature rise of a very small volume (fission track). A smooth spatial distribution of the heat production from fission is obtained from superposition of many fission tracks and by rapid heat transfer over the very short distance between the tracks. The range of fission products is so small compared to typical fuel rod diameters that almost all of the energy of the fission fragments is deposited in the fuel.

A smaller part of the total energy per fission (~11%) consists of practically “instantaneous” gamma rays and the energy directly deposited by neutrons in elastic scattering. “Instantaneous” in this context means that the energy is deposited with an insignificant delay, i.e., during about one neutron generation time after its production. The “instantaneous” gamma-ray energy results directly from the fission process, from inelastic neutron scattering or from radiative capture. As the mean free paths of neutrons are much larger than the fuel rod diameter, this part of the energy per fission is produced in all components of the core and even transported into the reflector or blanket. Still, the largest part of it is produced in the fuel – either directly from fission or from inelastic scattering and radiative capture processes as both reactions occur predominantly in the fuel. Furthermore, since the fuel has the highest reaction cross-section² from gamma rays in the core, a large portion of this gamma-ray energy is also deposited in the fuel.

The third part of the fission energy, the decay heat appears with a delay that is significant in dynamics. As a result of these delays and of the transport of some of the energy out of the fuel, ~10% less energy per fission is deposited in the fuel during a rapid transient than during a steady-state operation. That fraction is either directly deposited in the cladding and coolant or set free with delay in the form of beta particles and gamma rays at the time of fission products beta decays.

The conversion of the number of fissions into the various components of heat must reflect the distinctions discussed above. If C_{fsh} presents the heat per fission in a stationary system where fsh indicates “**f**ission energy $\rightarrow$ **s**tationary **h**eat”, the stationary total power P_0^{total} is written as:

$$P_0^{total} = C_{fsh} \int_V \int_E \Sigma_{f0} \phi_0 dE dV, \quad (II.13)$$

with

$$C_{fsh} \approx 0.35 \times 10^{-10}, \text{ W s/fission.} \quad (II.14)$$

These equations include the contribution of gammas from neutron capture and the delay energy release.

The power increase during a rapid transient, however, must be calculated with a different conversion factor, C_{fph} , which describes the rapid or **p**rompt conversion of **f**ission into **h**eat. If

² The cross-sections of Compton effect, pair production, and photo effect increase with Z, the number of protons, as Z, Z², and Z⁴, respectively.

C_{fdh} describes the production of **delayed heat** (through decay of fission products), the following balance exists:

$$C_{fsh} = C_{fph} + C_{fdh} . \quad (\text{II.15})$$

Typical values obtained from experimental investigations of the fission process are:

$$\frac{C_{fph}}{C_{fsh}} \approx 0.94 \quad (\text{II.16a})$$

and

$$\frac{C_{fdh}}{C_{fsh}} \approx 0.06 . \quad (\text{II.16b})$$

During transients, only the heat promptly produced needs to be considered. In very slow transients, a portion of the delayed heat also has to be considered, which complicates the treatment of these transients. A further complication results from the explicit treatment of the heat that is directly deposited in the cladding and coolant through gamma rays or neutrons. The gross space distribution of the heat deposition by gammas and neutrons differs from that of fission fragments and beta particles. The treatment of these complications is beyond the scope of this text. Only the prompt heat production in the fuel will be treated explicitly. The conversion factor C_{fhf} denotes the corresponding constant, which converts the number of **fission** into promptly available **heat** in the **fuel**. The promptly deposited heat amounts to 90% of the stationary heat. Thus, the conversion constant C_{fhf} has the value

$$C_{fhf} \approx 0.9 \cdot C_{fsh} . \quad (\text{II.17})$$

The remaining 10% of the heat production, which is delayed and/or is deposited outside of the fuel rods, is not explicitly considered in this chapter because of the emphasis on dynamics with prompt reactivity feedback.

Safety considerations and accident analysis are concerned primarily with the behavior of the power in the reactor core. The ~3 to 10% contribution that leaves the core in the form of neutrons or gammas or that is produced by fission in a fast breeder reactor (FBR) blanket normally has no effect on an accident progression. Thus, in this chapter, $P(t)$ denotes the power of the core.

Generally, transients that start from a stationary state are considered. The power transient $P(t)$ can then be split into the stationary power P_0 and the additional power $\delta P(t)$:

$$P(t) = P_0 + \delta P(t) , \quad (\text{II.18})$$

with

$$P_0 = C_{fsh} \int \int_{\text{core}} \Sigma_{f0} \phi_0 dE dV \quad (\text{II.19a})$$

and

$$\delta P(t) = C_{fhf} \int \int_{\text{core}} (\Sigma_f \phi - \Sigma_{f0} \phi_0) dE dV . \quad (\text{II.19b})$$

The distinction between prompt and delayed heat release is to be considered in the calculation of the feedback effects for rapid transients.

The increase in the fuel temperature results from the additionally produces fission energy. Subsequent heat transfer leads to a delayed increase of cladding and coolant temperature; this increase is in addition to the instantaneous and delayed energy deposition through gamma rays and neutrons. Since the transfer of heat out of material with low heat conductivity such as UO_2 - PuO_2 is fairly slow process, simplified heat transfer models are used in this chapter.

During rapid transients, most of the energy deposited in the fuel raises the fuel temperature and is not transferred to the cladding or coolant. In survey calculations, the heat transfer out of the fuel, therefore, can be treated by simple approximations; for example, it can be partially neglected. The neglect of fuel heat transfer, i.e., the adiabatic boundary condition of thermodynamics at the fuel cladding boundary, can be applied either to all of the heat or only to the incrementally produced heat. This results in the following two formulas for the average fuel temperature rise, $\delta T(t)$, after the onset of a transients (the effect of the temperature space distribution is below discussed):

$$\delta T(t) = C_{QT} \int_0^t P(t') dt' \begin{cases} \text{completely} \\ \text{adiabatic fuel} \end{cases} \quad (\text{II.20})$$

or

$$\delta T(t) = C_{QT} \int_0^t [P(t') - P_0] dt' \begin{cases} \text{adiabatic fuel for} \\ \text{power increment} \end{cases}, \quad (\text{II.21})$$

where P denotes the core power; C_{QT} , the conversion factor between energy and temperature, is below calculated.

The approximation, Eq. (II.20), is the same as Eq. (II.21) if the initial power is negligible compared to the power during the transient. Equation (II.20) was applied in the Manhattan Project. This approximate model of complete neglect of heat transfer is often referred to as the “linear energy model”.

The adiabatic model for the incremental energy, Eq. (II.21), is based on the following argument: The heat current out of the fuel rod is proportional to the negative gradient of the temperature distribution. The distribution of the additionally produced heat is essentially flat over the rod; thus, during a rapid transient, the temperature distribution is merely lifted up without any significant change in the shape of the distribution.

The adiabatic model can be improved in a first approximation by treating the heat release out of the fuel with a single time constant, λ_H :

$$\delta Q_{\text{remaining in fuel}} = \int_0^t [P(t') \exp[-\lambda_H(t - t')]] dt', \quad (\text{II.22a})$$

$$\delta Q_{\text{released from fuel}} = \int_0^t \delta P(t') \{1 - \exp[-\lambda_H(t - t')]\} dt', \quad (\text{II.22b})$$

and

$$\delta Q_{\text{total}} = \int_0^t \delta P(t') dt'. \quad (\text{II.22c})$$

The heat release constant λ_H for a cylindrical fuel rod of radius R with uniform heat production rate and temperature-independent heat conductivity is proportional to $1/R^2$:

$$\lambda_H \propto \frac{1}{R^2}. \quad (\text{II.23})$$

For a 6-mm (~ 0.25 -in)-diameter oxide fuel rod, one has

$$\lambda_H \approx \frac{1}{2s}. \quad (\text{II.24})$$

The energy-temperature conversion constant for the fuel $C_{QT}[K/MW \cdot s]$ in Eqs. (II.20) and (II.21) is given by

$$C_{QT} \left[\frac{K}{MW \cdot s} \right] = \frac{1}{c_h} \left[\frac{K \cdot mm^3}{W \cdot s} \right] \frac{1}{V_f [mm^3]} \cdot \frac{W}{MW}, \quad (\text{II.25})$$

where V_f is the fuel volume. The fuel heat capacity per unit volume, c_h , is given by the product of the specific heat, c_p , and density, ρ_d :

$$c_h \left[\frac{W \cdot s}{K \cdot mm^3} \right] = c_p \left[\frac{W \cdot s}{K \cdot g} \right] \rho_d \left[\frac{g}{mm^3} \right]. \quad (\text{II.26})$$

The conversion factor C_{QT} presents the rise of the average fuel temperature in the core per $MW \cdot s$ heat energy. Thus, if δQ_{cf} is the energy deposited in the core fuel, the corresponding temperature increase is given by:

$$\delta T(t) = C_{QT} \delta Q_{cf}(t). \quad (\text{II.27})$$

According to Eq. (II.25), C_{QT} is inversely proportional to the fuel volume, whereas δQ_{cf} is proportional to the fuel volume for a given rise in the specific power, or in the power per unit fuel volume. Equation (II.27) can then be rewritten as:

$$\delta T(t) = C_{QT} V_f \frac{\delta Q_{cf}(t)}{V_f}, \quad (\text{II.28a})$$

$$\delta T(t) = \frac{1}{c_h} \left[\frac{K \cdot mm^3}{W \cdot s} \right] \delta q_{cf} \left[\frac{W \cdot s}{mm^3} \right], \quad (\text{II.28b})$$

and

$$\delta T(t) = C_{qT} \delta q_{cf}(t), \quad (\text{II.28c})$$

where q_{cf} is the energy *density* deposited in the fuel and C_{qT} is the corresponding conversion factor.

For the calculation of $\delta T(t)$ it is convenient to introduce a separation of power into the nominal power, P_n (usually called “full power”), and the *flux* amplitude function $p(t)$,

$$P(t) \approx P_n p(t), \quad (\text{II.29})$$

The “ $\approx$ ” sign needs to be used in Eq. (II.29) since $p(t)$ describes only approximately the amplitude of the power. In this approximation then, p_0 represents the ratio of initial to nominal power. Inserting this separation into Eq. (10.15) with δQ taken from Eq. (II.22c) gives

$$\delta T(t) = \frac{1}{c_h} \frac{P_n[W]}{V_f[mm^3]} \int_0^t \delta p(t') dt'. \quad (II.30)$$

The integral over the amplitude difference, $\delta p(t) = p(t) - p_0$, was introduced earlier as:

$$I(t) = \int_0^t [p(t') - p_0] t'. \quad (II.31)$$

The integral $I(t)$ describes the additional energy in dimensionless units, which are called here “relative full-power seconds”. For example, $I(t) = 1.5$ gives an energy release of 1.5 full-powers second (fp-s). The conversion factor in Eq. (II.30) is denoted by C_{IT} , in analogy with other notations:

$$\delta T(t) = C_{IT} I(t). \quad (II.32)$$

The important conversion factor C_{IT} has dimension of K/s ; it is the rate of temperature rise per second of (uncooled) nominal power application:

$$C_{IT}[K/\text{fp-s}] = \left(\frac{\partial T}{\partial t} \right)_{P_n}. \quad (II.33)$$

Note that “fp-s” has a dimension of time and not a dimension of energy as the product $P_n \cdot \text{second}$ or $\text{fp} \cdot \text{s}$:

$$C_{IT}[K/s] = C_{IT}[K/\text{fp-s}]. \quad (II.34)$$

For example, for oxide fuel with an operational power density of 0.13 or 1.3 MW/l of fuel (0.13 or 1.8 W/mm³), typical of boiling water reactors (BWRs) or FBRs, respectively:

$$C_{IT} = \frac{1}{0.0032} \left[\frac{K \cdot mm^3}{W \cdot s} \right] \cdot \begin{cases} 0.13 \text{ W/mm}^3 \\ 1.3 \text{ W/mm}^3 \end{cases}$$

$$C_{IT} \approx \begin{cases} 40 \text{ K/fp-s} \\ 400 \text{ K/fp-s} \end{cases}; \quad (II.35)$$

i.e., 40 or K per full-power second. For pressurized water reactors (PWRs), C_{IT} is about twice as large as for BWRs.

The relations presented above hold for the average fuel temperature in the core, which was denoted by T for simplicity. The fuel temperature within a fuel rod varies strongly around this average. The fuel rod averaged temperature also depends on space. The “average” core fuel temperature, T , was used above as basis for the calculation of prompt reactivity feedback.

For the fairly flat power distribution across the fuel rods, the temperature distribution within the rod is approximately given by a parabola in the radial coordinate r :

$$T(r) = T_0 - \frac{r^2}{R^2}(T_0 - T_R), \quad (\text{II.36})$$

where R designates the fuel radius, and T_0 and T_R the temperatures at $r=0$ and $r=R$, respectively. The rod average value, $\bar{T}_{rod}$, can be obtained from Eq. (II.36) as:

$$\bar{T}_{rod} = \frac{1}{2}(T_0 + T_R). \quad (\text{II.37})$$

Equation (II.36) is derived under the assumption of temperature-independent thermodynamic properties and a space-independent heat production rate within a fuel rod: the latter assumption is only applied in survey investigations. The temperature variation within the fuel rod described by Eq. (II.36) is quite significant due to the high power density and due to the low heat conductivity of the oxide fuel. In fast reactors, $(T_0 - T_R)$ may be larger than 1000 K across an oxide fuel rod, but only about one-tenth of it for metal fuel.

The Doppler reactivity feedback effect is generally calculated for media with a uniform temperature distribution and not for the strongly space-dependent temperature profile within a fuel rod. An effective temperature therefore must be found that yields the correct Doppler reactivity when it is inserted into the formula based on a uniform temperature distribution. The effect of rod temperature profile on the Doppler effect of neutrons was first investigated for individual resonances and for $1/T$ variation of a local Doppler reactivity effect. Even though the analytical formulas that are derived in both investigations are different due to different physical assumptions, the numerical results are essentially in agreement for temperatures corresponding to a normal power operation. Both investigations yield an effective temperature that is ~3% smaller than the fuel rod average temperature. A correction of this type can be easily applied in practical calculations. In this text, the linear averaged temperature $T(=\bar{T})$ is used for simplicity.

The temperature distribution in the core, i.e., the r and z dependence of the rod averaged temperatures, could also be approximated by an effective “average” in analogy to that in the fuel rod. The reactivity effect for “isothermal” cores, $\delta\rho(T_{iso})$, would have to be determined such that effective “average” temperature change gives the same $\delta\rho$ that would be obtained from the gross temperature distribution in the core.

In more recent investigations, the Doppler feedback reactivity for the gross temperature distribution, $T(r, z)$, is calculated and a detailed feedback reactivity, $\delta\rho[T(r, z)]$ (normally in the form of a regional dependency), is applied directly instead of an isothermal quantity. Consequently, an effective temperature is unnecessary. In space-energy dependent dynamics, the reactivity effect of the local temperature is described by temperature-dependent group constants.

Prompt Feedback

There is only one really prompt feedback effect in thermal as well as in fast reactors, the Doppler reactivity effect. The corresponding feedback loop is as short as possible since the Doppler broadening of resonances occurs simultaneously with the dissipation of the kinetic energy of the fission products into temperature.

A reactivity insertion, $\delta\rho_{in}$, causes a change in power, $\dot{P}(t)$. The time integration of the kinetics equation yields $\delta P(t)$. The subsequent accumulation of energy, $\delta Q(t)$, changes the temperature,

$\delta T(t)$, which in turn leads to a change in reactivity, $\delta\rho_{Dop}(t)$, that feeds back to $\delta\rho_{in}$. This is illustrated in Fig. II-5.

The Doppler reactivity effect is negative in thermal and in large fast reactors. It represents the most important inherent shutdown mechanism and thus helps to assure the nuclear reactors safety.

The axial expansion of the fuel is the other feedback effect that may be considered as “prompt” in most transients. Axial expansion causes a reduction in the average fuel density in the core due to an elongation of the fuel rods. This then leads to a decrease in the reactivity. The reactivity feedback through axial expansion is not as prompt as the Doppler feedback since a temperature rise does not directly cause a reactivity feedback; δT is only the basis of a force. Two further time integrations are required to obtain the axial displacement that causes the reactivity feedback, as illustrated in Fig. II-5.

The displacement is established almost at the speed of sound in the fuel material (typically ~ 1000 m/s); the corresponding delay is then in ~ 1 ms. This delay can be neglected in all thermal and fast reactor transients. The only exceptions are in disassembly investigations, such as cases where transients are eventually shut down by the displacement of fuel. The delay of the displacements and thus the difference of time behavior of feedback through the Doppler effect versus expansion plays a characteristic role in such transients. For all transients discussed here, the axial expansion is considered a prompt feedback effect and appears added to the Doppler feedback.

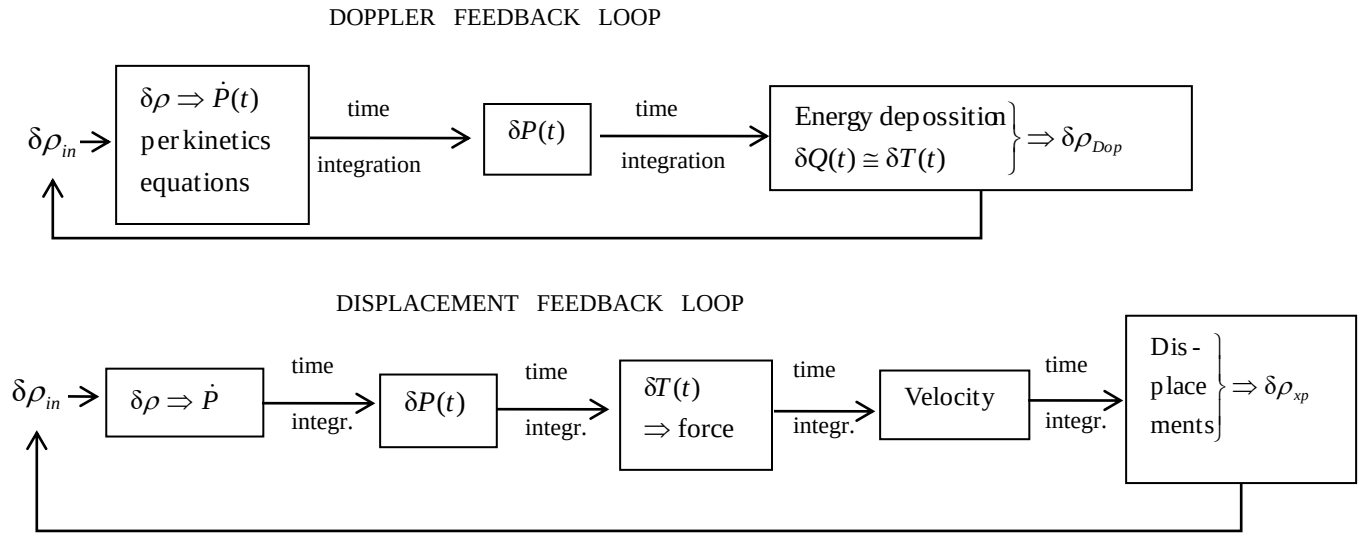


Fig. II-5. A schematic of Doppler and displacement feedback loops.

The reactivity effect of the Doppler broadening of resonances is calculated in the form of a temperature coefficient. For the calculation of feedback reactivity acting in rapid transients, it is convenient to convert the temperature coefficient into an energy coefficient by using one of the temperature-energy conversion formulas, Eq. (II.27) or (II.32):

$$C_{QT} \left(\frac{\partial \rho}{\partial T} \right)_{Dop} = \gamma_e^D \left[\frac{\delta \rho}{\text{MW} \cdot \text{s}} \right] = \text{energy coefficient} , \quad (\text{II.38a})$$

$$C_{IT} \left(\frac{\partial \rho}{\partial T} \right)_{Dop} = \gamma^D \left[\frac{\delta \rho}{\text{fp} \cdot \text{s}} \right] = \text{full - powersecond coefficient} . \quad (\text{II.38b})$$

The latter formula is used predominantly here since the transients are described by the flux amplitude $p(t)$. If the power as such is used, Eq. (II.38a) should be applied to describe the feedback.

Analogous formulas hold for the energy coefficient due to axial fuel expansion, γ^{exp} , in either of the units of Eqs. (II.38). Normally, the different time characteristics of the Doppler and expansion feedback loops need not be distinguished. Then, both energy coefficients appear as a sum:

$$\gamma = \gamma^D + \gamma^{\text{exp}}. \quad (\text{II.39})$$

If the fuel is treated adiabatically for the power increment and if the temperature dependence of the energy coefficient is neglected, the feedback reactivity is simply given by

$$\delta\rho(t) = \gamma \int_0^t [p(t') - p_0] dt'. \quad (\text{II.40})$$

Inclusion of first-order heat transfer yields:

$$\delta\rho(t) = \gamma \int_0^t [p(t') - p_0] \exp[-\lambda_H(t - t')] dt'. \quad (\text{II.41})$$

The temperature dependence of the Doppler coefficient, $(\partial\rho/\partial T)_D$, shows approximately a $1/T^x$ behavior:

$$\left(\frac{\partial\rho}{\partial T} \right)_D = \frac{A_D}{T^x}. \quad (\text{II.42})$$

For thermal reactors, x is close to 0.5 and for fast reactors, close 1.0.

The expansion coefficient is essentially independent of temperature. The temperature dependence of the sum of two coefficients results primarily from the Doppler effect. If the temperature dependence of the temperature coefficient is taken into account, feedback reactivity in the simple case of adiabatic approximation for the increment heat is calculated from

$$\delta\rho(t) = \int_{T_0}^{T(t)} \frac{\partial\rho}{\partial T} dT = \int_0^t \frac{\partial\rho}{\partial T} \frac{\partial T}{\partial t'} dt', \quad (\text{II.43})$$

with

$$\frac{\partial T}{\partial t'} = C_{\pi} [p(t') - p_0]. \quad (\text{II.44})$$

If the Doppler coefficient is proportional to $1/T$, as in large oxide-fueled fast reactors, a “Doppler constant”, $A_D = T(\partial\rho/\partial T)_D$, can be introduced. Its values are in the range of

$$0.003 \leq |A_D| \leq 0.009. \quad (\text{II.45a})$$

Division by an approximate operating reactor average fuel temperature of $t \approx 1500 \text{ K}$ gives

$$2 \times 10^{-6} / \text{K} \leq \left| \left(\frac{\partial\rho}{\partial T} \right)_D \right| \leq 6 \times 10^{-6} / \text{K}. \quad (\text{II.45b})$$

The value of A_D and $T(\partial\rho/\partial T)_D$ for a sodium-cooled fast reactors are in the upper part of the ranges given in Eq. (II.45); the corresponding values for gas-cooled fast reactors are a factor of ~ 2 smaller than for sodium-cooled reactors.

In the numerical fast reactor examples presented below, the following coefficient is used without specifying precisely the contributions of Doppler effect and expansion or the value of β :

$$\gamma_T = \frac{\partial\rho}{\partial T} = -0.002\$/K \text{ for FBRs,} \quad (\text{II.46a})$$

Doppler coefficients in thermal reactors with low fuel enrichment are generally much larger than in fast reactors by about one order of magnitude. This is due to the fact that the cross-section resonances that strongly contribute to Doppler feedback (the ones around and below neutron energy of several keV) are exposed to a higher portion of the neutron spectrum in thermal than in fast reactors. Thus,

$$\gamma_T = \frac{\partial\rho}{\partial T} = -0.02\$/K \text{ for LWRs,} \quad (\text{II.46b})$$

Using $C_{IT} = 400 \text{ K/fp-s}$ or 40 K/fp-s , Eq.(II.35), with Eqs.(II.46), one obtains

$$\gamma = C_{IT} \left(\frac{\partial\rho}{\partial T} \right) = -0.8\$/\text{fp-s} . \quad (\text{II.47})$$

Since the value of the feedback coefficient γ given in Eq. (II.47) is fairly typical for FBR and LWR oxide-fueled power reactors, it is used throughout the examples and semi-quantitative evaluations presented here. The two values for C_{IT} of Eq.(II.35) are applied for simplicity to distinguish LWR from FBR transient results.

Transients in the Subprompt-Critical or Subcritical Reactivity Domain

For simplicity, the initially inserted reactivity is approximated in this section by a step, ρ_1 . The treatment of transients following a subprompt reactivity step is greatly simplified by employing some of the approximations discussed in the previous sections. Even with such simplifications, however, a complete analytical solution has not yet been found. To obtain a semi-quantitative understanding of this important class of transients, the flux at small times, the asymptotic flux behavior, and the time integral, i.e., the total energy release, are investigated. The approximate results obtained from these investigations are compared with complete calculations (normally using six groups of delayed neutrons) presented in the graphs of this chapter.

In many of the approximate evaluations in this chapter, the prompt jump approximation (PJA) is applied. Though most comparisons with complete calculations are for the small Λ values of fast reactors ($\Lambda \approx 4 \times 10^{-7} \text{ s}$), the typical Λ values for LWRs ($\Lambda \approx 10^{-5} \text{ s}$) are small enough to allow the neglect of $\Lambda \dot{p}(t)$. Thus, the PJA is generally a good approximation on reactivity domain $\rho < \beta$ (better, $\rho \leq 0.9\beta$).

Since the feedback coefficients γ measured in dollars per full-power second are similar in magnitude for all oxide-fueled power reactors, the results that depend primarily on γ are applicable to LWRs and FBRs as well.

Transient at Small Times

The one-delay-group kinetics provides a good approximation for small times ($t \approx 0.2 / \bar{\lambda}$). The point kinetics equations with feedback reactivity are given by (ρ_1 denotes the initial step reactivity):

$$\Lambda \dot{p} = \left\{ \rho_1 + \gamma \int_0^t [p(t') - p_0] dt' - \beta \right\} p + \sum_k \lambda_k \zeta_k \quad (\text{II.48a})$$

and

$$\dot{\zeta}_k = -\lambda_k \zeta_k + \beta_k p. \quad (\text{II.48b})$$

The PJA (neglect of $\Lambda \dot{p}$) is employed to further simplify the problem. This leads to the approximate kinetics equation for the flux,

$$0 = \left\{ \rho_1 - \beta + \gamma \int_0^t [p(t') - p_0] dt' \right\} p + \sum_k \lambda_k \zeta_k \quad (\text{II.49a})$$

which is solved with the initial condition

$$p(0) = p^0 = \frac{\beta}{\beta - \rho_1} p_0. \quad (\text{II.49b})$$

Transients analyzed with the PJA may “start” with a pseudo-initial flux p^0 , which eliminates the rapid flux change during the prompt jump. The flux variations after the prompt jump are comparatively slow; they can be neglected by a Taylor series for small t . Note that without the PJA a Taylor expansion at $t=0$ would result in a tedious description of the prompt jump and would not allow a description of $p(t)$ after the prompt jump with only a few terms.

The competitive action of the increasing delayed neutron source and the negative prompt feedback is already exhibited in the second term of the Taylor expansion. Thus, only the first two terms of the Taylor expansion of p , ρ , and ζ are investigated in the following:

$$\begin{aligned} p(t) &= p^0 + p' t + \dots \\ \rho(t) &= \rho_1 + \rho' t + \dots \\ \zeta_k(t) &= \zeta_{k0} + \zeta'_k t + \dots \end{aligned} \quad (\text{II.50})$$

With $\delta\rho(t)$ from Eq. (II.40), one obtains for $\dot{p}$:

$$\begin{aligned} \dot{p} &= \gamma [p(t) - p_0] \\ &= \gamma [p^0 - p_0] + \dots \end{aligned} \quad (\text{II.51})$$

Inserting the approximate expansions into Eq. (II.48b) gives the rate of increase of the precursor density immediately after the prompt jump as

$$\begin{aligned} \zeta'_k &= -\lambda_k \zeta_{k0} + \beta_k p^0 \\ &= \beta_k (p^0 - p_0) \end{aligned} \quad (\text{II.52a})$$

From this rate of precursor increase, the following rate of increase of delayed neutron source results:

$$s'_d = (p^0 - p_0) \sum_k \beta_k \lambda_k = (p^0 - p_0) \beta \bar{\lambda} . \quad (\text{II.52b})$$

The precursor density and thus the delayed neutron source increase linearly right after the prompt jump. In the PJA, the increase in the delayed neutron source causes an instantaneous linear increase in the flux level. On the other hand, the instantaneous counteraction of the prompt feedback tends to decrease the flux. Let p'_{del} and p'_{fb} denote the two parts of the total slope that correspond to delayed neutrons and prompt feedback, respectively:

$$p' = p'_{del} + p'_{fb} . \quad (\text{II.53})$$

Both parts can be obtained by inserting the Taylor expansion into Eq. (II.49a); using s'_d from Eq. (II.52b) gives:

$$0 = [\rho_1 - \beta + \gamma(p^0 - p_0)t](p^0 - p't) + \beta p_0 + (p^0 - p_0) \beta \bar{\lambda} t . \quad (\text{II.54})$$

At $t=0$, this equation reduces to

$$(\rho_1 - \beta)p^0 + \beta p_0 = 0 , \quad (\text{II.55})$$

which is satisfied by the definition of p^0 , Eq.(II.49b). The slope is obtained from the terms proportional to t , which are given by:

$$[\rho_1 - \beta]p' + \gamma(p^0 - p_0)p^0 + (p^0 - p_0)\beta \bar{\lambda} = 0 . \quad (\text{II.56})$$

Omitting the feedback yields the contribution of the increasing precursor population to the slope; Eq. (II.56) gives for $\gamma = 0$,

$$p'_{del} = \bar{\lambda} \frac{\beta}{\beta - \rho_1} (p^0 - p_0) . \quad (\text{II.57a})$$

Inserting $p^0 = \beta p_0 / (\beta - \rho_1)$ from Eq. (II.49b) brings p'_{del} into the form

$$p'_{del} = \bar{\lambda} \frac{p^0}{p_0} (p^0 - p_0) . \quad (\text{II.57b})$$

Thus, the linear *increase* of the flux due to the increase in the delayed neutron source is proportional to the product of flux and flux jump.

Omitting the increase of precursors in Eq. (II.56) yields the influence of feedback on the variation of the flux after the prompt jump:

$$\begin{aligned} p'_{fb} &= \gamma \frac{(p^0 - p_0)}{\beta - \rho_1} p^0 \\ &= \frac{\gamma}{\beta} \frac{p^0}{p_0} (p^0 - p_0) p^0 \end{aligned} \quad (\text{II.58})$$

Thus, the counteraction of prompt reactivity feedback leads to a linear *decrease* of the flux, which contains one further flux factor, p^0 , compared to p'_{del} , since $\dot{p}$ is influenced by the product of $\delta\rho \cdot p$.

Adding p'_{del} and p'_{fb} and bracketing out the common factor gives

$$p' = \left[\bar{\lambda} + \frac{\gamma}{\beta} p^0 \right] \frac{p^0}{p_0} (p^0 - p_0) \quad (\text{II.59})$$

which shows that for small p^0 the slope p' can be (and practically is) positive, but for large prompt jumps, the feedback starts to turn the flux down immediately after the prompt jump.

For one special prompt jump, p^{00} , the linear flux change after the prompt jump vanishes. From Eq. (II.59) follows

$$-\frac{\gamma}{\beta} p^{00} = \bar{\lambda} \quad (\text{II.60})$$

If p is considered as the power, then Eq. (II.60) can be interpreted as the influence of feedback and delayed neutrons on the flux change after the prompt jump is cancelled, e.i., when the energy coefficient times the prompt jump power is equal to the average precursor delay constant.

Subprompt-critical transients of this type can be used to measure Doppler coefficient in fast reactor. Figure II-6 shows a computer simulation of transients introduced by a delayed supercritical reactivity step around the special transient for which the slope after the prompt jump is practically zero. The lower curve ($\rho_1 / \beta = 0.940$) is still rising after prompt jump; the upper curves in Fig. II-6 are clearly decreasing immediately after prompt jump. Around $\rho_1 / \beta \approx 0.945$, the slope is practically zero after the prompt jump. The corresponding value of p^{00} is ~ 18 times the initial flux. The energy coefficient obtained from Eq. (II.60) is:

$$\frac{\gamma}{\beta} = -\frac{\bar{\lambda}}{p^{00}} \quad (\text{II.61})$$

With $\bar{\lambda} \approx 0.40$ (^{235}U was used as fuel and $P_0 = 10 \text{ MW}$ in this simulation) and $p^{00} \approx 18p_0$, Eq. (II.61) yields

$$\gamma \approx -0.022 \text{ } \$/\text{fp} - \text{s} \quad (\text{II.62a})$$

or

$$\gamma \approx -2.2 \times 10^{-3} \frac{\$}{\text{MW} \cdot \text{s}} \quad (\text{II.62b})$$

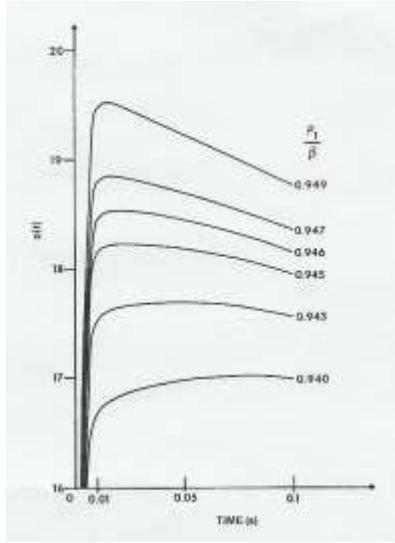


Fig. II-6. Delayed supercritical ρ -step-induced transients.

The value 2.0×10^{-3} , which is obtained from the simulated analysis, is 2% different from the value of $2.15 \times 10^{-3} \frac{\$}{\text{MW} \cdot \text{s}}$

actually used to calculate the flux transient of Fig. II-6 from the kinetics equations with six delayed groups. This comparison also shows the accuracy of the PJA with a single delayed group in a transient feedback. The value of γ , Eq. (II.62a), is ~ 35 times lower than that given in Eq. (II.47) since the power density in this computer simulation is ~ 16 times smaller and the β of the ^{235}U fuel is about two times larger than in a fast power reactor.

Asymptotic Transients

In addition to the expansion at $t = 0$, Eq. (II.50), an asymptotic expansion of Eq. (II.49a) can be derived. It follows from such a derivation that Eq. (II.49a) has a constant asymptotic (“as”) solution. With $\dot{p} = 0$, the kinetic equation in the PJA, assumes the form

$$0 = \frac{\lambda \rho_{as} + \dot{\rho}_{as}}{\beta - \rho_{as}} . \quad (\text{II.63a})$$

This requires that the numerator vanish, i.e.,

$$\lambda \rho_{as} + \dot{\rho}_{as} = 0 . \quad (\text{II.63b})$$

The asymptotic time-independent solution of Eq. (II.63b) is

$$\rho_{as} = 0 \text{ since } \dot{\rho}_{as} = 0 . \quad (\text{II.64})$$

The approach to the asymptotic solution above is describes by the time-dependent solution of Eq. (II.63b), i.e.,

$$\rho_{as}(t) \propto \exp(-\lambda t) . \quad (\text{II.65})$$

Since $\rho_{as} = 0$, the inserted reactivity is asymptotically compensated for by the feedback:

$$-\gamma \int_0^\infty [p(t') - p_0] dt' = \rho_1 , \quad (\text{II.66})$$

in which the integral has to converge, which requires $p(t)$ to converge sufficiently fast toward p_0 . Thus, a feedback reactivity given by the left side of Eq. (II.66) requires as the asymptotic flux:

$$P_{as} = P_0 . \quad (\text{II.67})$$

Under the assumptions, on which Eqs. (II.49) are based; the prompt feedback forces the power back to its initial value.

The fact that the asymptotic flux, p_{as} , with feedback described by the left side of Eq. (II.66), cannot be smaller than p_0 can be shown by the following physical argument. If $p(t) < p_0$ in a certain time interval, the contribution of this interval to the integral is negative. This physically means that the fuel temperature is decreasing. Decreasing fuel temperature together with a negative temperature coefficient leads to a *reactivity increase*, which is followed by an increase in the flux.

Consider as an example the highest of the curves in Fig. II-6 ($P_0=10$ MW). According to Eq. (II.67) and (II.66), the transient is supposed to converge to p_0 and its integral is approximately given by

$$\int_0^\infty [p(t') - p_0] dt' = -\frac{\rho_1}{\gamma} \left(= \frac{0.949}{2.15 \times 10^{-3}} \text{ MW} \cdot \text{s} \cong 430 \text{ MW} \cdot \text{s} \right), \quad (\text{II.68})$$

Figures II-7 and II-8 show two additional examples calculated with a typical power reactor energy coefficient, $\gamma = -0.8 \text{ \$/fp} \cdot \text{s}$ [see Eq. (II.47)]. The dashed lines show two transients for $\rho_1 = 0.95 \text{ \$}$ and $\rho_1 = 0.5 \text{ \$}$ with reactivity feedback given in Eq. (II.49a). The energy releases according to Eq. (II.68) are 1.19 and 0.625 fp-s, respectively. The dashed lines of Figs. II-7 and II-8 cross the initial flux levels at ~ 3 s and the flux approaches the asymptotic value, p_0 , from below. This overshooting of the final flux level is required as can be seen from the following consideration. The stationary flux level is related to the delayed neutron source and β by the source multiplication formula:

$$p_0 = \frac{s_{d0}}{\beta}.$$

This relation is certainly fulfilled initially, and it must also hold asymptotically since $p_{as} = p_0$. If $p(t) \geq p_0$ and $\rho(t) \geq 0$ during the entire transient, the reactor would be permanently “loaded” with additional precursors. The above source multiplication equation would then require the asymptotic flux to be larger than p_0 in contradiction to Eq. (II.67). To resolve this contradiction, the flux must be $< p_0$ for some period of time to eliminate the additional precursors that are produced in the early part of the transient in order to achieve asymptotically $s_d = s_{d0}$.

Equations (II.63) are based on these assumptions:

1. the PJA
2. one-delay-group kinetics
3. neglect of transient heat transfer.

The first assumption should be accurate asymptotically since it is violated only in the immediate vicinity of fast changes.

The use of one-delay-group kinetics might be questionable since the kinetics parameters cannot be lumped uniquely into a one-delay-group λ . The one-delay-group λ , however, does not even

appear in the asymptotic relations (II.67) and (II.68). Therefore, replacing a λ value by a different one that might asymptotically be more appropriate cannot improve the results in Eqs. (II.67) and (II.68). The reason is that the asymptotic state is a stationary state and therefore the flux level is independent of the precursor decay constant altogether.

The third assumption, the neglect of heat transfer, is violated during the transient and even more so asymptotically. The influence of heat transfer on the asymptotic flux can be semi-quantitatively investigated by using Eq. (II.41). With $\rho_{as} = 0$, the following equation is obtained instead of Eq. (II.66):

$$\lim_{t \rightarrow \infty} -\gamma \int_0^t [p(t') - p_0] \exp[-\lambda_H(t - t')] dt' = \rho_1 . \quad (\text{II.69})$$

Asymptotically, p_{as} can be inserted as the value of the flux amplitude in performing the limit to large t . The exponential function practically eliminates the contribution of the integrand, which is far away from the upper limit. Asymptotically, the lower limit “0” is infinitely far away from “t”; it can therefore be replaced by $-\infty$. This gives:

$$-\gamma \int_{-\infty}^t [p_{as} - p_0] \exp[-\lambda_H(t - t')] dt' = \rho_1 , \quad (\text{II.70})$$

or

$$-\frac{\gamma}{\lambda_H} (p_{as} - p_0) = \rho_1 . \quad (\text{II.71})$$

Solving Eq. (II.71) for the asymptotic flux amplitude yields:

$$p_{as} = p_0 - \rho_1 \frac{\lambda_H}{\gamma} > p_0 . \quad (\text{II.72})$$

Thus, the prompt negative feedback does *not* shut the reactor *down* in sense of making it subcritical with an asymptotically vanishing flux. Therefore, the control or shutdown system must be employed to restore the initial power or shut the reactor down. Only if p_0 is practically zero and λ_H is very small, i.e., if there is practically no heat release, can p_{as} be very small. An example of an uncooled core with transient starting at $p_0 = 0$ is provided by the Godiva experiments. Figures II-7 and II-8 show as solid lines two transients ($\rho_1 = 0.95\%$ and $\rho_1 = 0.5\%$) as they develop with feedback and including first-order heat transfer ($\lambda_H = 0.5 \text{ s}^{-1}$).

The basic effect of heat transfer is that the flux is not forced back to its initial value. It assumes asymptotically a value above the initial flux. The numerical calculation of the transient yields the same asymptotic flux value as Eq. (II.72). The formula based on a single delay constant λ_H provides the correct qualitative understanding of why an asymptotically constant flux above the initial flux is obtained.

Independent of the simplifications employed in describing the heat transfer, the negative reactivity feedback restores criticality. The inserted reactivity ρ_1 is compensated by a temperature rise:

$$\Delta T = -\rho_1 \left(\frac{\partial \rho}{\partial T} \right)^{-1}. \quad (\text{II.73})$$

With $\partial \rho / \partial T \approx -0.002 \text{ \$}/\text{K}$ [Eq. (II.46)], where T is the average fuel temperature, a 0.5\$ (or 0.95\$) reactivity insertion would lead to a temperature rise of 250 K (or 475 K) if the control system did not start its counteraction during the transient. The temperature rise in the center of the hottest pellet may be substantially larger than that given by Eq. (II.73).

The following is a summary of basic conclusions for model transients in power reactors that can be drawn from the results presented in this chapter:

1. Prompt negative reactivity feedback reduces a subprompt reactivity insertion to zero: i.e., it restores criticality and does *not shut* the reactor *down*.
2. A permanently inserted reactivity is compensated by a *permanent temperature rise*, given by Eq. (II.73).
3. The flux approaches asymptotically a constant value. An asymptotic power level higher than the original power is required to sustain a permanent temperature rise.
4. If a core is originally at near zero power and “uncooled”, the Doppler feedback resulting from a “permanent” temperature increase will reduce the neutron flux to near zero and lead to a “shutdown” in this limited sense. However, the asymptotic core would still be critical.

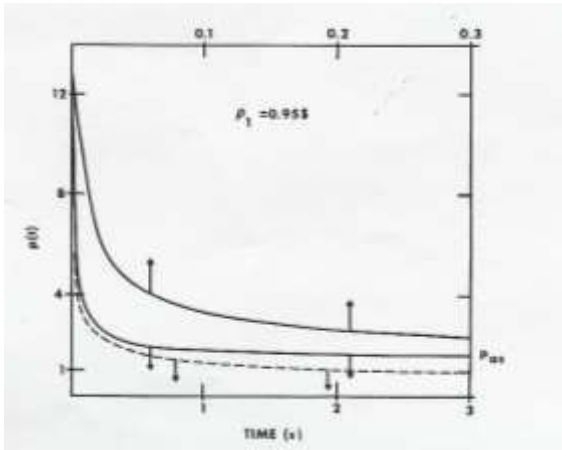


Fig. II-7. Subprompt-critical transient ($\rho_1=0.95$ \$, $p_0=1$) with reactivity feedback including first-order heat transient (solid line) and no heat transfer (dashed line).

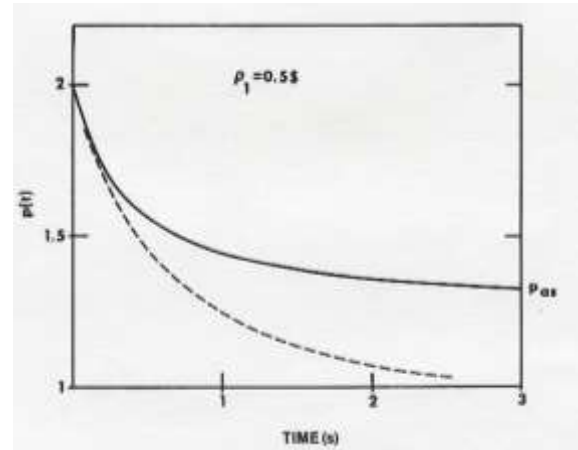


Fig. II-8. Subprompt-critical transient ($\rho_1=0.5$ \$, $p_0=1$) with reactivity feedback including first-order heat transient (solid line) and no heat transfer (dashed line).

Review Questions

- 1a. What are energy, temperature, and power coefficients?
- b. Give examples of physical phenomena involved.

- c. Give an example of a transient for the application of energy, temperature, and power coefficients, respectively.
- 2a. Give the formula for the adiabatic feedback model for the entire and the incremental energy respectively.
- b. Give a typical application for each of the two models.
- 3. Describe the general concepts of the use of reactivity coefficients and their application (for constant and variable coefficients).
- 4. Describe qualitatively the transient at small times (including the two competing effects). Which approximation allows an accurate quantitative evaluation of those effects?
- 5a. Derive (as briefly as possible) the “slope” of the flux as it results from the increased delayed neutron population.
- b. Derive (as briefly as possible) the “slope” of the flux as it results from feedback.
 - c. Compare both slopes and find the reactivity insertion at which they compensate each other.
- 6. Find the value of the asymptotic flux (derived from the assumption that it is constant): (a) without heat transfer and (b) with heat transfer described by a single exponential function.